

UnicodeEncodeError: 'charmap' codec can't encode characters

Asked 8 years, 4 months ago Modified 4 days ago Viewed 918k times



473



I'm trying to scrape a website, but it gives me an error.



I'm using the following code:



```
import urllib.request
from bs4 import BeautifulSoup

get = urllib.request.urlopen("https://www.website.com/")
html = get.read()

soup = BeautifulSoup(html)

print(soup)
```

And I'm getting the following error:

```
File "C:\Python34\lib\encodings\cp1252.py", line 19, in encode
    return codecs.charmap_encode(input,self.errors,encoding_table)[0]
UnicodeEncodeError: 'charmap' codec can't encode characters in position 70924-70950:
character maps to <undefined>
```

What can I do to fix this?

[python](#) [beautifulsoup](#) [urllib](#) [Edit tags](#)

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asked Nov 23, 2014 at 18:47



[SstrykerR](#)

7,502 3 12 11

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81



In Python 3.7, and running Windows 10 this worked (I am not sure whether it will work on other platforms and/or other versions of Python) Replacing this line:

```
with open('filename', 'w') as f:
```





With this:

```
with open('filename', 'w', encoding='utf-8') as f:
```

The reason why it is working is because the encoding is changed to UTF-8 when using the file, so characters in UTF-8 are able to be converted to text, instead of returning an error when it encounters a UTF-8 character that is not support by the current encoding.

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edited Mar 22 at 18:44

answered Oct 8, 2018 at 16:20



Ian Thompson

2,814 2 18 30



Sabbir Ahmed

985 6 7

-
- 2 ▲ print(soup) return \xd0\xbf\xd0\xbe\xd0\xb6\xd0\xb0\xd0\xbb\xd1\x83\xd0\xb9\xd – Coffee inTime
 Nov 28, 2019 at 19:26
-
- 1 ▲ @CoffeeinTime That looks like UTF-16 erroneously converted to some 8-bit encoding, or possibly using Pyhon 2. The string you show is truncated, but it seems to begin with "뽕뽕뽕뽕뽕뽕" (I don't read Korean so I have no idea if that makes any sense). Demo: ideone.com/092Jnk – tripleee
 Jul 20, 2022 at 9:37
-



2

I got the same error so I use (encoding="utf-8") and it solve the error. This generally happens when we got some unidentified symbol or pattern in text data that our encoder does not understand.



```
with open("text.txt", "w", encoding='utf-8') as f:
    f.write(data)
```

This will solve your problem.

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answered Nov 19, 2022 at 19:11



Shubham rawat

59 2



682

I was getting the same `UnicodeEncodeError` when saving scraped web content to a file. To fix it I replaced this code:



```
with open(fname, "w") as f:
    f.write(html)
```



with this:

```
with open(fname, "w", encoding="utf-8") as f:
    f.write(html)
```

If you need to support Python 2, then use this:

```
import io
with io.open(fname, "w", encoding="utf-8") as f:
    f.write(html)
```

If you want to use a different encoding than UTF-8, specify whatever your actual encoding is for `encoding`.

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edited Oct 13, 2022 at 18:04

answered Feb 27, 2017 at 21:00



twasbrillig

16.4k 9 42 64

-
- 21 ▲ In mac(python 3) works perfectly with just open without encoding, but in windows(w10, python3) is not an option. Just works in that way, with `encoding="utf-8"` param. – [xtornasol512](#) Apr 30, 2017 at 1:14
-
- 1 ▲ OP requested to read the file however, not write the file. The issue seems to be console-related. – [NaturalBornCamper](#) Sep 20, 2019 at 14:01
-



5



There are multiple aspects to this problem. The fundamental question is which character set you want to output into. You may also have to figure out the input character set.

Printing (with either `print` or `write`) into a file with an explicit `encoding="..."` will translate Python's internal Unicode representation into that encoding. If the output contains characters which are not supported by that encoding, you will get an `UnicodeEncodeError`. For example, you can't write Russian or Chinese or Indic or Hebrew or Arabic or emoji or ... anything except a restricted set of some 200+ Western characters to a file whose encoding is `"cp1252"` because this limited 8-bit character set has no way to represent these characters.

Basically the same problem will occur with any 8-bit character set, including nearly all the legacy Windows code pages (437, 850, 1250, 1251, etc etc), though some of them support some additional script in addition to or instead of English (1251 supports Cyrillic, for example, so you can write Russian, Ukrainian, Serbian, Bulgarian, etc). An 8-bit encoding has only a maximum of 256 character codes and no way to represent a character which isn't among them.

Perhaps now would be a good time to read Joel Spolsky's [The Absolute Minimum Every Software Developer Absolutely, Positively Must Know About Unicode and Character Sets \(No Excuses!\)](#).

On platforms where the terminal is not capable of printing Unicode (only Windows these days really, though if you're into retrocomputing, this problem was also prevalent on other platforms in the previous millennium) attempting to `print` Unicode strings can also produce this error, or output [mojibake](#). If you see something like `HÃ©llö` instead of `Héllö`, this is your issue.

In short, then, you need to know:

If you are on Windows and write UTF-8 to a text file, maybe specify `encoding="utf-8-sig"` which adds a BOM sequence at the beginning of the file. This is strictly speaking not necessary or correct, but some Windows tools need it to correctly identify the encoding.

Several of the earlier answers here suggest blindly applying some encoding, but hopefully this should help you understand how that's not generally the correct approach, and how to figure out - rather than guess - which encoding to use.

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edited Sep 8, 2022 at 13:51

answered Jun 23, 2022 at 15:47



tripleee

171k

31

263

310



Perhaps see also meta.stackoverflow.com/questions/379403/... – tripleee Aug 1, 2022 at 18:16



From Python 3.7 onwards, Set the the environment variable `PYTHONUTF8` to 1

2

The following script included other useful variables too which set System Environment Variables.



```
setx /m PYTHONUTF8 1
setx PATHEXT "%PATHEXT%;.PY" ; In CMD, Python file can be executed without extesnion.
setx /m PY_PYTHON 3.10 ; To set default python version for py
```

[Source](#)

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answered Sep 5, 2022 at 4:52



Smart Manoj

4,945

4

32

58

- 2 The code snippet seems to be for Windows. For Unix systems with a Bourne-family shell, try `export PYTHONUTF8=1` (though probably this type of hack is only realistically necessary on Windows anyway). – tripleee Sep 8, 2022 at 13:46



55



You may or may not need to set that second environment variable [PYTHONLEGACYWINDOWSTDIO](#).



Alternatively, this can be done in code (although it seems that doing it through env vars is recommended):

```
sys.stdin.reconfigure(encoding='utf-8')
sys.stdout.reconfigure(encoding='utf-8')
```

Additionally: **Reproducing this error** was a bit of a pain, so leaving this here too in case you need to reproduce it on your machine:

```
set PYTHONIOENCODING=windows-1252
set PYTHONLEGACYWINDOWSTDIO=windows-1252
```

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answered Aug 25, 2020 at 7:06



Voy

4,932 1 47 57

- 1 This is perfect; I was getting this error while using the Python Debugger (pdb) on a Windows system looking at source code that used utf-8 and had lots of emoji in it. Every time I did a "list" command to see where I was, the "charmap" error appeared. Settings these two environment variables made my debugging as smooth as silk. – [nutjob](#) Oct 9, 2020 at 18:31
- 4 `sys.stdin.reconfigure` is invalid on Python 3.9.0, it throws `AttributeError: 'StdInputFile' object has no attribute 'reconfigure'` – [Suncatcher](#) Dec 24, 2020 at 10:13
- 6 On Windows 10, using GIT BASH, setting the env variables mentioned above did NOT work, however, setting the two lines in the actual python code file DID work:

```
sys.stdin.reconfigure(encoding='utf-8') sys.stdout.reconfigure(encoding='utf-8')
```

– [Henrik Carlström](#) Apr 12, 2021 at 9:39
- @Suncatcher Try to run this Python script in a different IDE – [Petr L.](#) Aug 15, 2021 at 19:38
- @PetrL. why I should use IDE at all? all valid Python commands should be interpretable in Python Shell, otherwise they are not valid – [Suncatcher](#) Aug 16, 2021 at 10:33

|



-1



if you are using windows try to pass **encoding='latin1', encoding='iso-8859-1' or encoding='cp1252'** example:

```
csv_data = pd.read_csv(csvpath,encoding='iso-8859-1')
print(print(soup.encode('iso-8859-1')))
```



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answered Jul 31, 2020 at 22:16



Karim Sherif

399 2 5



I fixed it by adding `.encode("utf-8")` to `soup`.

258

That means that `print(soup)` becomes `print(soup.encode("utf-8"))`.



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edited Dec 4, 2019 at 19:36



twasbrillig

16.4k 9 42 64

answered Nov 23, 2014 at 19:19



SstrykerR

7,502 3 12 11



-
- 7 ▲ don't hardcode the character encoding of your environment (e.g., console) inside your script, [print Unicode directly instead](#) – [jfs](#) Sep 7, 2015 at 4:09
-
- 5 ▲ This is just printing the repr of a `bytes` object, which will print as a mess of `\x` sequences if there's a lot of UTF-8 encoded text. I recommend using `win_unicode_console`, as [@J.F.Sebastian](#) suggests. – [Eryk Sun](#) May 23, 2016 at 20:48
-
- 7 ▲ This makes it print out `b'\x02\xxc2\xa9'` (a bytes object) instead – [MilkyWay90](#) Jun 16, 2019 at 16:20
-



14



Even I faced the same issue with the encoding that occurs when you try to print it, read/write it or open it. As others mentioned above adding `.encoding="utf-8"` will help if you are trying to print it.

```
soup.encode("utf-8")
```

If you are trying to open scraped data and maybe write it into a file, then open the file with `(.....,encoding="utf-8")`

```
with open(filename_csv , 'w', newline='',encoding="utf-8") as csv_file:
```

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answered May 16, 2019 at 16:30

[Pardhu Gopalam](#)

179 1 6



21



While saving the response of get request, same error was thrown on Python 3.7 on window 10. The response received from the URL, encoding was UTF-8 so it is always recommended to check the encoding so same can be passed to avoid such trivial issue as it really kills lots of time in production

```
import requests
resp = requests.get('https://en.wikipedia.org/wiki/NIFTY_50')
print(resp.encoding)
with open ('NiftyList.txt', 'w') as f:
    f.write(resp.text)
```

When I added `encoding="utf-8"` with the open command it saved the file with the correct response

```
with open ('NiftyList.txt', 'w', encoding="utf-8") as f:
    f.write(resp.text)
```

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edited Dec 22, 2018 at 14:36

answered Dec 22, 2018 at 14:31



Suraj Rao

29.4k

11

96

103



Abhishek Jain

3,637

2

24

25

For those still getting this error, adding `encode("utf-8")` to `soup` will also fix this.

5

```
soup = BeautifulSoup(html_doc, 'html.parser').encode("utf-8")
print(soup)
```

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edited Nov 5, 2018 at 22:01

answered Oct 1, 2016 at 19:09



MilkyWay90

1,983

1

8

20



Pseudo Sudo

1,392

3

15

34

12 `soup` is no longer a `BeautifulSoup` object after you do this so it cannot be manipulated or searched – [NaturalBornCamper](#) Sep 20, 2019 at 13:53

1 This is already adequately covered by both the OP's own answer, and by Pardhu Gopalam's. – [Karl Knechtel](#) Sep 5, 2022 at 4:49



This post is hidden. It was [deleted](#) last year by [7uc1f3r](#), [Mark Rotteveel](#), [Zoe stands with Ukraine](#) .

-2

Thankyou so much, I worked for me

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answered Mar 26, 2022 at 6:59



Aravind R

577

1

5

13

2 Please don't add "thank you" as an answer. Instead, vote up the answers that you find helpful. - [From Review](#) – [Gino Mempin](#) Mar 26, 2022 at 8:05

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-1

If you are on windows run **chcp 65001** in the cmd prompt.

Then run your code :)

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answered Oct 13, 2020 at 17:24



Tanish Sarmah

430

5

14

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-3



This post is hidden. It was [deleted](#) 6 years ago by the post author.

Another way I've always used this is to decode the data before putting it into soup. So essentially, the line to read html become `html = get.read().decode('UTF-8')`.

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answered Nov 2, 2016 at 13:12



Jialu

114 10

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0



This post is hidden. It was [deleted](#) 7 years ago by [Robert Harvey](#).

I am facing the same problem, but with this message:

```
Traceback (most recent call last):
File "ELD.py", line 83, in <module>
    BeginSession() # Begin the session
File "ELD.py", line 72, in BeginSession
    CreateAccount()
File "ELD.py", line 23, in CreateAccount
    complete_process = input("You are creating the account \"" + desired_username + "\"
with the password " + ("\u2022" * len(desired_password)) + ". Are you sure you want to
do this [Y/N]: ")
File "C:\Users\Aanand Kainth\AppData\Local\Python\lib\encodings\cp437.py", line 12, in
encode
    return codecs.charmap_encode(input,errors,encoding_map)
UnicodeEncodeError: 'charmap' codec can't encode characters in position 58-65:
character maps to <undefined>
```

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answered Feb 5, 2016 at 18:49



Aanand Kainth

121 1 1 6



This does not really answer the question. If you have a different question, you can ask it by clicking [Ask Question](#). You can also [add a bounty](#) to draw more attention to this question once you have enough [reputation](#). - [From Review](#) – [Lafexlos](#) Feb 5, 2016 at 19:42

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-1



This post is hidden. It was [deleted](#) 8 years ago by [Blazemonger](#), [Amro](#), [John Saunders](#).

Will not let me upvote, but had the same problem and SstrykerR second reply with the `".encode("utf-8")"` worked like a charm!

Thanks.

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answered Mar 17, 2015 at 19:54



Trojandnc

1

1 This does not provide an answer to the question. To critique or request clarification from an author, leave a comment below their post - you can always comment on your own posts, and once you have sufficient [reputation](#) you will be able to [comment on any post](#). – honk Mar 17, 2015 at 20:05

Please don't add "thank you" as an answer. Once you have sufficient [reputation](#), you will be able to [vote up questions and answers](#) that you found helpful. – Amro Mar 17, 2015 at 23:20

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